

Supplementary Material for “Mitigating Digital Twin-enabled Electrochemical Degradation in Optimal Ship-to-Grid Dispatch for Nordic Nearshore Energy Islands”

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I. EFFECT OF SEI THICKNESS ON ITS RATE OF GROWTH

In Fig. 1, we plot $\Delta\theta_i^{t+1}$ with respect to S_i^t and I_i^t given different values of θ_i^t . Via visual inspection, we find that the $\Delta\theta_i^t$ surfaces do not vary significantly across $\theta_i^t = 0$ nm, $\theta_i^t = 20$ nm, $\theta_i^t = 40$ nm, and $\theta_i^t = 80$ nm. This offers justification for neglecting the effects of θ_i^t in Θ in Section II-A of the main paper.

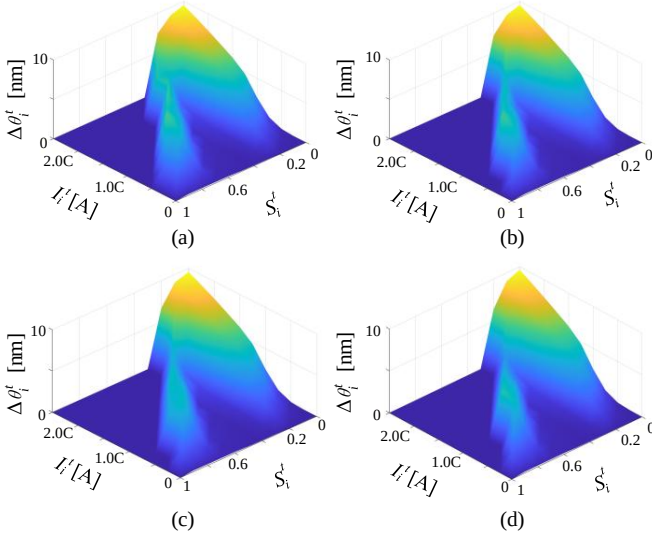


Fig. 1. PDE-based SEI growth surface $\Delta\theta_i^t$ with (a) $\theta_i^t = 0$ nm, (b) $\theta_i^t = 20$ nm, (c) $\theta_i^t = 40$ nm, and (d) $\theta_i^t = 80$ nm.

II. DERIVATION OF SECOND-ORDER CONIC CONSTRAINT

Define $f := (\gamma_i^{t+1} + 1)^2$ and $g := (\gamma_i^{t+1} - 1)^2 + (2z_{i,N_I}^{t+1})^2$. Then

$$f - g = 4\gamma_i^{t+1} - 4(z_{i,N_I}^{t+1})^2 \geq 0, \quad (\text{II-1})$$

where the inequality holds because $\gamma_i^{t+1} \geq (z_{i,N_I}^{t+1})^2$. Rearrangement of (II-1) yields

$$(\gamma_i^{t+1} + 1)^2 \geq (\gamma_i^{t+1} - 1)^2 + (2z_{i,N_I}^{t+1})^2, \quad (\text{II-2})$$

which is equivalent to the standard second-order conic constraint

$$\|\gamma_i^{t+1} + 1\|_2^2 \geq \|[\gamma_i^{t+1} - 1, 2z_{i,N_I}^{t+1}]^T\|_2^2. \quad (\text{II-3})$$

III. BATTERY CELL PARAMETER VALUES

For numerical case studies presented in Section III of the main paper, each UES is equipped with 21700-type battery cells in Fig. 2, with the capacity of 80 kWh. Each UES battery pack has 96 cells in series and 46 cells in parallel (96S46P), with a total of 4416 cells. Each battery cell has capacity 0.005 kWh, nominal voltage $v_{\text{flat},i} = 3.65$ V with operating voltage ranging from 2.5 V to 4.2 V, and maximum C-rate of 2.5 to avoid excessively sharp charge-discharge cycling. For a battery cell, $\bar{p}_i \approx 3.65 \cdot 0.005 \cdot 2.5 \approx 0.046$ kWh. The ambient temperature is assumed to be 28°C. Other parameters related to electrochemical dynamics are given in Table I.

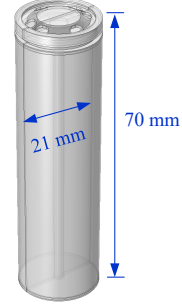


Fig. 2. A 21700-type battery cell in COMSOL.

IV. LINEAR BATTERY AGING COST FORMULATION

In [6], the linear battery degradation cost F_i^t is defined for each charging and discharging cycle at time t until the required state of charge (SoC) in the last time period $t = N_t$, i.e., $S_i^{N_t} = \bar{S}$, yielding

$$F_i^t = C_i \frac{\kappa_i}{100} \left(-\frac{1}{\eta_v} p_{d,i}^t \theta t + \bar{S} - S_i^0 \right) \quad (\text{IV-4})$$

where κ_i denotes the slope of the linear approximation of the UES i 's battery life as a function of cycles, C_i represents the purchase cost of the UES i 's battery divided by its useful capacity [\$/Ah], and S_i^0 is the initial SOC value.

V. COMPARISON WITH NONLINEAR BATTERY AGING COST FORMULATION

In [8], the nonlinear battery degradation cost G_i^t is obtained from experimental data as an approximation with respect to depth of discharge (DoD) $1 - S_i^t$ and cell current I_i^t (denoted in C-rate). We have

$$G_i^t = \frac{1}{f_c(I_i^t) \cdot f_d(S_i^t)}, \quad (\text{V-5})$$

where f_c and f_d represent battery life loss due to DoDs and cell currents, respectively, and they are given by

$$f_d = \begin{cases} 4000, & \text{if } S_i^t > 0.95, \\ 946.1 \cdot (1 - S_i^t)^{-1.079}, & \text{if } S_i^t \leq 0.95, \end{cases} \quad (\text{V-6a})$$

$$f_c = \begin{cases} 4, & \text{if } I_i^t < 0.2, \\ 1.041 \cdot (I_i^t)^{-0.445}, & \text{if } 0.2 \leq I_i^t \leq 2.5. \end{cases} \quad (\text{V-6b})$$

Substitution of the above in (V-5) then yields

$$G_i^t = \begin{cases} 0.6 \cdot 10^{-4}, & \text{if } S_i^t > 0.95, I_i^t < 0.2, \\ 0.2 \cdot 10^{-3} \cdot (I_i^t)^{0.445}, & \text{if } S_i^t > 0.95, 0.2 \leq I_i^t \leq 2.5, \\ 0.2 \cdot 10^{-3} \cdot (1 - S_i^t)^{1.079}, & \text{if } S_i^t \leq 0.95, I_i^t < 0.2, \\ 10^{-3} \cdot (1 - S_i^t)^{1.079} \cdot (I_i^t)^{0.445}, & \text{if } S_i^t \leq 0.95, 0.2 \leq I_i^t \leq 2.5. \end{cases} \quad (\text{V-7a})$$

TABLE I
PARAMETERS FOR ELECTROCHEMICAL KINETICS

Parameter	Negative electrode	Separator	Positive electrode
electrode plate area [m ²]	0.163	0.163	0.163
electrode thickness [m]	$78 \cdot 10^{-6}$	$20 \cdot 10^{-6}$	$45 \cdot 10^{-6}$
Li ⁺ diffusion coefficient [m ² /s]	$3.9 \cdot 10^{-5}$	-	$1.8 \cdot 10^{-8}$
active electrode volume fraction [%]	0.6	-	0.6
electrolyte phase volume fraction [%]	0.3	-	0.3
max solid phase concentration [mol/m ³]	31507	-	49000
particle radius [m]	$6 \cdot 10^{-6}$	-	$5 \cdot 10^{-6}$
reaction rate efficiency [A/m ²]	$9.77 \cdot 10^{-2}$	-	$1.19 \cdot 10^{-2}$
exchange current density of side reaction [A/m ²]	10	-	10
initial electrolyte concentration [mol/m ³]	$1.25 \cdot 10^4$	$1.25 \cdot 10^4$	$1.25 \cdot 10^4$
Binder volume fraction [%]	0.1	-	0.1
Separator volume fraction [%]	-	0.4	-
Thickness of current collectors [m]	$10 \cdot 10^{-6}$	-	$10 \cdot 10^{-6}$